

Advancement Analytics Data Model

Data Dictionary & Technical Reference

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Platform: Snowflake
Schemas: RAW (source tables) | ANALYTICS (views & models)

1. Overview

This data model consolidates constituent biographical data, institutional giving history, engagement records, and external philanthropic data into a unified Snowflake schema. It serves as the single source of truth for advancement reporting, prospect scoring, and major gift pipeline management.

The model powers two composite scores — Affinity Score (internal CRM data) and Propensity Score (external giving data) — which are combined to produce a Prospect Grade (A–D) for each constituent. This grade drives major gift officer assignment prioritization and eliminates the need for manual prospect re-research.

2. Schema Architecture

Layer	Object Type	Purpose
RAW	Tables	Source data as loaded from CRM and external vendor feeds. Minimally transformed.
ANALYTICS	Views	Transformed, aggregated, and scored layers built on top of RAW tables.
ANALYTICS	V_UNIFIED_PROSPECT_MODEL	Final reporting model — single row per constituent with all scores, grades, and assignment data joined.

3. Source Tables (RAW Schema)

RAW.CONSTITUENT

Core biographical and demographic record for each constituent. One row per person. Foundation for all other joins.

Column Name	Data Type	Nullable	Description
CONSTITUENT_ID	VARCHAR(10)	NO	Primary key. Unique identifier assigned by CRM system.
FIRST_NAME	VARCHAR(50)	YES	Legal first name as recorded in CRM.
LAST_NAME	VARCHAR(50)	YES	Legal last name as recorded in CRM.
FULL_NAME	VARCHAR(100)	YES	Computed column: FIRST_NAME ' ' LAST_NAME.
CLASS_YEAR	INTEGER	YES	Year of graduation or expected graduation.
DEGREE_TYPE	VARCHAR(20)	YES	Degree classification: 'UG', 'GRAD', or 'CERT'.
CONSTITUENT_TYPE	VARCHAR(20)	YES	Relationship to institution: 'Alumni UG', 'Alumni Grad', 'Friend', 'Parent', 'Past Parent'.

Column Name	Data Type	Nullable	Description
PRIMARY_EMAIL	VARCHAR(100)	YES	Primary contact email address.
CITY	VARCHAR(50)	YES	City of primary residence.
STATE	VARCHAR(2)	YES	Two-character US state code.
IS_DECEASED	BOOLEAN	NO	Excludes record from all active prospect pools when TRUE.
IS_DISQUALIFIED	BOOLEAN	NO	Excludes record from prospect scoring when TRUE (e.g. staff, faculty).
RESEARCH_RATING	VARCHAR(30)	YES	Vendor wealth rating label (e.g. '\$100,000–249,999').
RESEARCH_RATING_AMT	NUMBER(15,2)	YES	Numeric version of research rating for range filtering and sorting.
NU_AFFILIATION_FLAG	BOOLEAN	NO	TRUE if constituent has active NU club, board, or committee affiliation.
CREATED_AT	TIMESTAMP_NTZ	NO	Record creation timestamp.
UPDATED_AT	TIMESTAMP_NTZ	NO	Last modification timestamp.

RAW.NU_GIFT

All hard-credit gift transactions made to Northeastern University. One row per gift. Used to calculate Affinity Score components.

Column Name	Data Type	Nullable	Description
GIFT_ID	VARCHAR(20)	NO	Primary key. Unique transaction identifier.
CONSTITUENT_ID	VARCHAR(10)	NO	Foreign key to RAW.CONSTITUENT.
GIFT_DATE	DATE	YES	Date the gift was received or pledge was made.
FISCAL_YEAR	INTEGER	YES	NU fiscal year (July–June). Derived from GIFT_DATE.
GIFT_AMOUNT	NUMBER(15,2)	YES	Gift amount in USD.
GIFT_LEVEL	VARCHAR(20)	YES	'Annual' (\$1–\$999), 'Leadership' (\$1,000–\$9,999), 'Major' (\$10,000+).
FUND_CODE	VARCHAR(20)	YES	Designation fund code from CRM.
FUND_NAME	VARCHAR(100)	YES	Human-readable fund name.
GIFT_TYPE	VARCHAR(20)	YES	'Outright', 'Pledge', or 'Bequest'.
IS_HARD_CREDIT	BOOLEAN	NO	TRUE = gift counts toward donor's record. Soft credits excluded from scoring.

RAW.ENGAGEMENT

Events, personal visits, volunteer activities, and declined opportunity records. Used to calculate visit and affiliation components of the Affinity Score.

Column Name	Data Type	Nullable	Description
ENGAGEMENT_ID	VARCHAR(20)	NO	Primary key.
CONSTITUENT_ID	VARCHAR(10)	NO	Foreign key to RAW.CONSTITUENT.
ENGAGEMENT_TYPE	VARCHAR(30)	YES	'Personal Visit', 'Event', 'Volunteer', 'Board', 'Declined Opp'.
ENGAGEMENT_DATE	DATE	YES	Date of the engagement activity.
FISCAL_YEAR	INTEGER	YES	NU fiscal year mapped from ENGAGEMENT_DATE.
STAFF_NAME	VARCHAR(100)	YES	Name of the MGO or staff member who logged the engagement.
NOTES	VARCHAR(500)	YES	Free-text notes from the engagement record.

RAW.EXTERNAL_GIVING

Philanthropic giving to non-NU organizations, sourced from external wealth screening vendors. Used exclusively to calculate Propensity Score. High/low gift ranges are used instead of exact amounts due to vendor data privacy practices.

Column Name	Data Type	Nullable	Description
EXT_GIFT_ID	VARCHAR(20)	NO	Primary key.
CONSTITUENT_ID	VARCHAR(10)	NO	Foreign key to RAW.CONSTITUENT.
ORGANIZATION_NAME	VARCHAR(200)	YES	Name of the recipient organization.
ORG_CATEGORY	VARCHAR(50)	YES	'Higher Ed', 'Non-Profit', 'Religious', 'Other'.
LOW_GIFT_AMOUNT	NUMBER(15,2)	YES	Lower bound of the reported gift range.
HIGH_GIFT_AMOUNT	NUMBER(15,2)	YES	Upper bound of the reported gift range. Used in scoring calculations.
GIFT_YEAR	INTEGER	YES	Calendar year the external gift was made.
IS_NONPROFIT	BOOLEAN	NO	TRUE if the recipient organization is a registered nonprofit.

RAW.PROSPECT_ASSIGNMENT

Major gift officer portfolio assignments and prospect pipeline stage tracking. Populated after scoring model output is reviewed by the research team.

Column Name	Data Type	Nullable	Description
ASSIGNMENT_ID	VARCHAR(20)	NO	Primary key.
CONSTITUENT_ID	VARCHAR(10)	NO	Foreign key to RAW.CONSTITUENT.
ASSIGNED_TO	VARCHAR(100)	YES	Name of the assigned Major Gift Officer.
ASSIGNMENT_DATE	DATE	YES	Date the prospect was formally assigned.
STATUS	VARCHAR(20)	YES	'Active', 'Inactive', or 'Prospect' (unassigned but in pool).

Column Name	Data Type	Nullable	Description
STAGE	VARCHAR(30)	YES	Pipeline stage: 'Identification', 'Cultivation', 'Solicitation', 'Stewardship'.
PROSPECT_GRADE	VARCHAR(2)	YES	Model-derived grade written back after scoring: 'A', 'B', 'C', or 'D'.

4. Analytics Views (ANALYTICS Schema)

ANALYTICS.V_NU_GIVING_SUMMARY

Aggregates all hard-credit NU gift transactions into one row per constituent. Intermediate layer consumed by the Affinity Score view.

Field Name	Data Type	Description / Derivation
CONSTITUENT_ID	VARCHAR	Foreign key from RAW.CONSTITUENT.
TOTAL_GIFT_COUNT	INTEGER	Total number of hard-credit NU gifts.
GIVING_YEARS	INTEGER	Number of distinct fiscal years with at least one gift.
LIFETIME_GIVING	NUMBER	Sum of all hard-credit NU gift amounts.
LARGEST_GIFT	NUMBER	Maximum single gift amount.
AVG_GIFT	NUMBER	Mean gift amount across all transactions.
MOST_RECENT_FY	INTEGER	Most recent fiscal year in which a gift was made.
CAMPAIGN_GIVING_YEARS	INTEGER	Count of fiscal years with giving during FY2010–FY2016 campaign window.
HAS_MAJOR_GIFT	INTEGER	1 if constituent has at least one Major-level gift (\$10,000+); else 0.
HAS_LEADERSHIP_GIFT	INTEGER	1 if constituent has at least one Leadership-level gift (\$1,000–\$9,999); else 0.
RECENT_GIVER_FLAG	INTEGER	1 if constituent gave in FY2015 or FY2016; else 0.
CONSISTENCY_SCORE	INTEGER	0 = gave <3 campaign years; 1 = gave 3–4 years; 2 = gave 5+ years.

ANALYTICS.V_ENGAGEMENT_SUMMARY

Aggregates all engagement activity into one row per constituent. Feeds visit and recency components of the Affinity Score.

Field Name	Data Type	Description / Derivation
CONSTITUENT_ID	VARCHAR	Foreign key from RAW.CONSTITUENT.
TOTAL_ENGAGEMENTS	INTEGER	Total count of all engagement records.
TOTAL_VISITS	INTEGER	Count of Personal Visit engagements only.
RECENT_VISIT_FLAG	INTEGER	1 if a Personal Visit occurred in FY2014, FY2015, or FY2016; else 0.
MOST_RECENT_ENGAGEMENT_FY	INTEGER	Most recent fiscal year with any engagement on record.
ENGAGEMENT_YEARS	INTEGER	Number of distinct fiscal years with engagement activity.

ANALYTICS.V_EXTERNAL_GIVING_SUMMARY

Aggregates external vendor giving data into one row per constituent. Feeds all Propensity Score components.

Field Name	Data Type	Description / Derivation
CONSTITUENT_ID	VARCHAR	Foreign key from RAW.CONSTITUENT.
TOTAL_EXT_GIFTS	INTEGER	Total count of external gift records from vendor.
TOTAL_ORGS_SUPPORTED	INTEGER	Count of distinct organizations to which constituent has given.
MOST_RECENT_EXT_GIFT_YEAR	INTEGER	Most recent calendar year with an external gift.
LARGEST_EXT_GIFT	NUMBER	Highest gift HIGH_GIFT_AMOUNT across all external records.
HIGHER_ED_GIFT_COUNT	INTEGER	Count of gifts to Higher Ed organizations.
HAS_NONPROFIT_AFFILIATION	INTEGER	1 if any gift was to a nonprofit organization; else 0.
BROAD_GIVING_FLAG	INTEGER	1 if constituent has given to 5 or more distinct organizations; else 0.
HIGH_FREQUENCY_FLAG	INTEGER	1 if constituent has 10 or more total external gift records; else 0.
RECENT_EXT_GIVER_FLAG	INTEGER	1 if constituent made an external gift in 2011 or later; else 0.

ANALYTICS.V_AFFINITY_SCORE

Calculates the Affinity Score for each constituent using internal CRM data only. Scale: 0–13. Measures depth of connection and commitment to Northeastern University.

Field Name	Data Type	Description / Derivation
TOTAL_AFFINITY_SCORE	INTEGER	Sum of all component scores plus penalty. Primary output of this view.
AMOUNT_SCORE	INTEGER	0–3 based on highest NU gift level: Annual=1, Leadership=2, Major=3.
CONSISTENCY_SCORE	INTEGER	0–2 based on campaign giving years: <3 yrs=0, 3–4 yrs=1, 5+ yrs=2.
CONSTITUENT_SCORE	INTEGER	0–3 based on relationship type: UG Alumni=3, Grad Alumni=2, Parent=1, Other=0.
RECENCY_SCORE	INTEGER	1 if gave in FY2015 or FY2016; else 0.
VISIT_SCORE	INTEGER	1 if any personal visit on record; else 0.
RECENT_VISIT_SCORE	INTEGER	1 if personal visit occurred in FY2014–FY2016; else 0.
NU_AFFILIATION_SCORE	INTEGER	1 if active NU affiliation (club, board, committee); else 0.
DECLINED_OPP_PENALTY	INTEGER	-2 if any declined opportunity on record; else 0.

ANALYTICS.V_PROPENSITY_SCORE

Calculates the Propensity Score for each constituent using external giving data only. Scale: 0–8. Measures general philanthropic inclination independent of NU relationship.

Field Name	Data Type	Description / Derivation
TOTAL_PROPENSITY_SCORE	INTEGER	Sum of all component scores. Primary output of this view.
EXT_AMOUNT_SCORE	INTEGER	0–2 based on largest external gift: \$1,000–\$9,999=1, \$10,000+=2.
EXT_RECENCY_SCORE	INTEGER	1 if any external gift was made in 2011 or later; else 0.
EXT_SPREAD_SCORE	INTEGER	1 if constituent gave to 5 or more distinct organizations; else 0.
EXT_FREQUENCY_SCORE	INTEGER	1 if constituent has 10 or more total external gift records; else 0.
NONPROFIT_AFFILIATION_SCORE	INTEGER	1 if any external gift was to a nonprofit; else 0.
HIGHER_ED_BONUS_SCORE	INTEGER	0–2: Major HE gift=2, 10+ HE gifts=1, else 0. Rewards higher ed philanthropists.

ANALYTICS.V_UNIFIED_PROSPECT_MODEL

Final unified reporting model. One row per active, non-disqualified constituent. Joins all source tables, intermediate views, scores, and assignment data into a single queryable layer. This is the primary table consumed by the Prospect Tool and all downstream reporting.

Field Name	Data Type	Description / Derivation
CONSTITUENT_ID	VARCHAR	Unique constituent identifier. Primary key of this model.
FULL_NAME	VARCHAR	Computed full name from CONSTITUENT table.
CONSTITUENT_TYPE	VARCHAR	Relationship type: Alumni UG, Alumni Grad, Friend, Parent, Past Parent.
RESEARCH_RATING	VARCHAR	Vendor wealth rating label.
RESEARCH_RATING_AMT	NUMBER	Numeric wealth rating for range filtering.
NU_LIFETIME_GIVING	NUMBER	Sum of all hard-credit NU gifts.
CAMPAIGN_GIVING_YEARS	INTEGER	Years with NU giving during the FY2010–FY2016 Empower Campaign.
TOTAL_VISITS	INTEGER	Total personal visits on record.
EXT_GIFT_COUNT	INTEGER	Total external gift records from vendor.
EXT_ORGS_COUNT	INTEGER	Distinct external organizations supported.
TOTAL_AFFINITY_SCORE	INTEGER	Composite score measuring NU-specific connection. Scale 0–13.
TOTAL_PROPENSITY_SCORE	INTEGER	Composite score measuring general philanthropy. Scale 0–8.
COMBINED_SCORE	NUMBER	Average of Affinity and Propensity scores. Used for grade assignment.
PROSPECT_GRADE	VARCHAR	A/B/C/D grade derived from NTILE(4) ranking on COMBINED_SCORE. A = highest priority.
ASSIGNED_TO	VARCHAR	MGO name if prospect has been assigned; NULL if unassigned.
ASSIGNMENT_STATUS	VARCHAR	Current prospect status: Active, Inactive, Prospect, or NULL.

Field Name	Data Type	Description / Derivation
PROSPECT_STAGE	VARCHAR	Pipeline stage: Identification, Cultivation, Solicitation, Stewardship.

5. Included Analytical Queries

Query	Purpose
5A — Top 50 Unassigned Prospects	Returns the 50 highest-scoring unassigned prospects ranked by Combined Score. Primary input for MGO assignment decisions.
5B — Grade Distribution Summary	Aggregates prospect counts, average scores, and giving metrics by grade tier (A–D). Used for portfolio sizing and strategy planning.
5C — Giving Trend by Fiscal Year	Tracks total NU giving, unique donor count, and major/leadership gift volume across the FY2008–FY2017 campaign period.
5D — Constituent Type Performance	Compares participation rates, lifetime giving, and affinity scores across constituent types to identify highest-value relationship segments.
5E — High-Capacity Unvisited Prospects	Surfaces high-wealth prospects (rating \$500K+) with no visit on record and an A or B grade — a quick-win list for outreach planning.

6. Scoring Reference

Affinity Score Components (Internal CRM Data)

Component	Max Points	Scoring Logic
Gift Amount (NU Only)	3	Annual gift (\$1–\$999) = 1 Leadership (\$1K–\$9.9K) = 2 Major (\$10K+) = 3
Giving Consistency	2	Gave 3–4 campaign years = 1 Gave 5+ campaign years = 2
Constituent Type	3	UG Alumni = 3 Grad Alumni = 2 Parent = 1 Other = 0
Recency	1	Gave in FY2015 or FY2016 = 1
Visit	1	Has any personal visit on record = 1
Recent Visit	1	Had visit in FY2014–FY2016 = 1
NU Affiliation	1	Active NU club/board/committee affiliation = 1
Declined Opp Penalty	-2	Has any declined opportunity on record = -2
TOTAL AFFINITY SCORE	13	Sum of all components including penalty. Higher = stronger NU connection.

Propensity Score Components (External Giving Data)

Component	Max Points	Scoring Logic
External Gift Amount	2	Leadership external gift (\$1K–\$9.9K) = 1 Major (\$10K+) = 2
Recency	1	Made external gift in 2011 or later = 1
Spread (# Orgs)	1	Gave to 5 or more distinct organizations = 1
Frequency (# Gifts)	1	Has 10 or more total external gift records = 1

Component	Max Points	Scoring Logic
Nonprofit Affiliation	1	Has given to at least one nonprofit organization = 1
Higher Ed Bonus	2	Major HE gift (\$10K+) = 2 10+ HE gifts = 1 else 0
TOTAL PROPENSITY SCORE	8	Sum of all components. Higher = stronger general philanthropic behavior.

Prospect Grade Assignment

Grades are assigned using Snowflake's NTILE(4) window function, which divides the full prospect pool into four equal quartiles ranked by Combined Score (average of Affinity and Propensity). This ensures grades are always relative to the full pool rather than fixed score cutoffs.

Grade	Quartile	Interpretation
A	Top 25%	Highest-priority prospects. Immediate candidates for MGO assignment and major gift cultivation.
B	50th–75th	Strong prospects. Monitor for assignment as MGO capacity allows.
C	25th–50th	Moderate prospects. May warrant re-screening or additional research before assignment.
D	Bottom 25%	Lower priority. Retain in pool for future re-evaluation as circumstances change.